BITS Pilani

# Introduction to DevOps

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# CSI ZG514/SE ZG514 - Introduction to DevOps
# Lecture No. 4

# CS – 4 DevOps Dimensions

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# Coverage

- DevOps – People
- Team structure in a DevOps
- Transformation to Enterprise DevOps culture
- Building competencies, Full Stack Developers
- Self-organized teams, Intrinsic Motivation
- Technology in DevOps (Infrastructure as code, Delivery Pipeline, Release Management)
- Tools/technology as enablers for DevOps

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# DevOps – People

- Addresses the people aspect of adopting DevOps, including creating the necessary culture.

- An organization may adopt the most efficient processes or automated tools possible, but they're useless without the people who eventually must execute those processes and use those tools.

# DevOps – People

- DevOps CULTURE
- (1) Identifying business objectives
- Creating a culture is getting everyone headed in the same direction and working toward the same goal
- Identifying common business objectives for the team and the organization as a whole.
- identify goals that it wants to achieve; then it can develop a common set of milestones toward those goals for different teams of stakeholders to use
- Collaboration and communication across stakeholders

# DevOps – People

- (2) Create an environment of sharing

- leaders of the organization to work with their teams to create an environment and culture of collaboration and sharing.

- Leaders must remove any self-imposed barriers to cooperation.

- Typical measurements reward operations teams for uptime and stability, and reward developers for new features delivered, but they pit these groups against each other.

# DevOps – People

- (2) Create an environment of sharing

- The leaders of the organization should further encourage collaboration by improving visibility.

- Establishing a common set of collaboration tools is essential, especially when teams are geographically distributed and can't work together in person.

- Giving all stakeholders visibility into a project's goals and status is crucial for building a DevOps culture based on trust and collaboration.

# DevOps – People

- DEVOPS TEAM

- The arguments for and against having a separate DevOps team are as old as the concept itself.

- DevOps liaison teams, which resolve any conflicts and promote collaboration.

- Such a team may be an existing tools group or process group, or it may be a new team staffed by representatives of all teams that have a stake in the application being delivered.

- DevOps team, your most important goal is to ensure that it functions as a Center of Excellence that facilitates collaboration without adding a new layer of bureaucracy or becoming the team that owns addressing all DevOps related problems

# DevOps – People

## Key competencies to succeed

- Culture – Shared organizational assumptions to resolve problems
- Leadership
  - visioning, strategic management, flexibility, and the ability to inspire others to innovate and perform
  - individuals with advanced communications skills, the knowledge of diverse cultures, and people who behave collaboratively when working in teams

# DevOps – People

## Key competencies to succeed

- Resource/management
  - effective management of available resources
  - collaboration across multi-functional teams is at the heart of DevOps, delivering an accountable business outcome
- Subject matter experts (SME's) in different areas like – Automation, Build & Release management, CI/CD, Containers, Security, Cloud (AWS/GCP/Azure)
- DevOps engineers who are members from different department (development, QA, Operations )

# Create the ideal DevOps team structure

- A solid DevOps platform needs a solid DevOps team structure to achieve maximum efficiency.

- Building a robust DevOps team structure is not merely about assembling a group of individuals with technical skills; it's about fostering a DevOps culture that emphasizes collaboration, continuous improvement, and efficiency across the product lifecycle.

- As organizations look to streamline their software development process, understanding the roles of a DevOps engineer, the development team, and how automation tools can enhance productivity is vital.

- This holistic view helps in shaping a DevOps structure that aligns with the overarching goals of reliable software delivery and a productive work environment.

# Create the ideal DevOps team structure

- Several factors come into play when it comes to team structure:
- Existing silos: Are there product sets/teams that work independently?
- Technical leadership: Are group managers set up to achieve DevOps goals?
- Changing roles: Ops tasks have bled into dev roles, security teams are working with everyone, and technology is changing. Expect to regularly re-evaluate everything.
- Continuous improvement: A DevOps team will never be a “one and done.” Iteration will be required.

# Types of silos

- The division of Dev and Ops into separate teams often leads to challenges in the deployment process.
- However, embracing a DevOps culture where common tools are integrated can bridge these gaps.
- A few DevOps scenarios in great detail, but we'll discuss just a few of the silos specifically and how they impact an organization.

# Dev and ops are completely separate

- Skelton refers to this as a classic “throw it over the wall” team structure and, as implied, it’s not the most effective DevOps strategy.
- Both teams work in their bubbles and lack visibility into the workflow of the other team.
- This complete separation lacks collaboration, visibility, and understanding – vital components of what effective DevOps should be.
- What happens is essentially blame-shifting: “We don’t know what they are doing over there, we did our part and now it’s up to them to complete it,” and so on.

# DevOps middleman

- In this team structure, there are still separate dev and ops teams, but there is now a “DevOps” team that sits between, as a facilitator of sorts.
- This is not necessarily a bad thing and Skelton stresses that this arrangement has some use cases.
- For example, if this is a temporary solution with the goal being to make dev and ops more cohesive in the future, it could be a good interim strategy.

# DevOps middleman

- In this scenario, dev and DevOps are melded together while ops remains siloed.
- Organizations like this still see ops as something that supports the initiatives for software development, not something with value in itself.
- Organizations like this suffer from basic operational mistakes and could be much more successful if they understand the value ops brings to the table.

# What can DevOps team leadership do?

- To break down DevOps team silos requires leadership at all levels.
- Start by asking each group to surface the major areas of friction and then identify leaders in each group – dev, ops, security, test.
- Each leader should work individually and together on all of the friction points.
- The importance of communication can't be overstated: Teams need to hear regular feedback about all aspects of their roles.
- It might also be helpful to insert "champions" into struggling groups; they can model behaviors and language that facilitate communication and collaboration.

# DevOps roles are blurring

- Technology advances from multicloud to microservices and containers also play a role when it comes to defining the right DevOps team structure.

- In a 2020 Global DevSecOps Survey, 83% of respondents said their teams are releasing code more quickly but they also told us their roles were changing, dramatically in some cases.

- Devs today are creating, monitoring, and maintaining infrastructures, roles that were traditionally the province of ops pros.

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# DevOps roles are blurring

- Ops are spending more time managing cloud services, while security team members are working on cross-functional teams with dev and ops more than ever before.

- Obviously, the software development lifecycle today is full of moving parts, meaning that defining the right structure for a DevOps team will remain fluid and in need of regular re-evaluation.

# Remember to iterate

- We have a reliability group that manages uptime and reliability.
- A quality department, and a distribution team, just to name a few.
- The way that we make all these pieces fit together is through our commitment to transparency and our visibility through the entire SDLC.
- But we also tweak (i.e. iterate on) this structure regularly to make everything work.
- The bottom line: Plan to build your DevOps team, and then re-think it, and re-think it some more.
- The benefits in faster code releases and happier team members will make it worthwhile.

# Remember to iterate

- The journey to optimizing a DevOps team structure is iterative, reflecting the continual advancements in DevOps processes and tools.
- Each element plays a crucial role in the team's success.
- By breaking down traditional silos and integrating roles within DevOps teams, organizations can foster a more cohesive and efficient environment.
- Ultimately, the key to sustained improvement lies in regularly re-evaluating and refining the DevOps structure to keep pace with the fast-evolving demands of software production and deployment.
- This commitment not only speeds up the software development process but also builds a more resilient and responsive organization.

# How to lead a successful DevOps transformation (without burning out your teams)

- DevOps transformation has become essential for organizational survival.
- Yet despite the clear benefits, many transformation efforts fail to deliver expected results.
- Engineering leaders find themselves caught between executive demands for faster delivery and teams resistant to yet another disruptive change.
- The difference between successful transformations and failed experiments often comes down to leadership approach.
- This guide provides a practical blueprint for engineering directors and DevOps leaders who need to drive meaningful transformation without sacrificing team well-being.

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# What is DevOps transformation (and why it's harder than it sounds)?

- DevOps transformation is the systematic evolution of how you build, deploy, and operate software. It's not about installing new tools.
- It's about changing mindsets, collaboration models, and delivery rhythms.
- True transformation requires rewiring how your development and operations teams collaborate, breaking down traditional silos that separate those who build software from those who run it.
- This fundamentally shifts responsibilities, processes, and most critically, culture.

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# What makes DevOps transformation particularly challenging?

- Cultural resistance exceeds technical hurdles.
- Engineers comfortable with established workflows naturally resist changes that disrupt their routines.
- Mid-transformation chaos feels unavoidable.
- As you migrate from legacy to modern practices, you'll temporarily run parallel systems that can increase complexity.
- Leadership alignment is difficult but necessary.
- When executives and engineering leaders differ on transformation goals, teams receive mixed signals that hinder progress.

# What makes DevOps transformation particularly challenging?

- Measuring success isn't straightforward.
- Unlike pure technical projects, DevOps transformation success indicators blend technical metrics with cultural and business outcomes.
- The essence of DevOps extends far beyond continuous integration and deployment tools.
- It represents a fundamental shift in how teams collaborate, share responsibility, and deliver value.

# Why DevOps transformation is a business imperative (not just an IT project)

- DevOps transformation directly impacts business survival, not just technical efficiency.
- Companies that master DevOps practices consistently outperform their competitors across critical business metrics:
- **Revenue impact:** Companies with mature DevOps practices generate more revenue growth than their competitors.
- This stems from releasing features faster, responding to market changes quicker, and reducing lost revenue from outages.

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# Why DevOps transformation is a business imperative (not just an IT project)

- **Customer satisfaction:** When you deploy smaller changes more frequently, you reduce risk while delivering value continuously.

- This translates to more responsive product development and fewer customer-impacting issues.

- **Innovation capacity:** By automating routine work and reducing toil, DevOps practices free your engineers to focus on strategic innovations rather than maintenance and firefighting.

- **Talent acquisition and retention:** Engineers increasingly evaluate potential employers based on their technical practices.

- Organizations with modern DevOps approaches attract and retain top talent more effectively.

# Why DevOps transformation is a business imperative (not just an IT project)

- **Competitive resilience:** Market demands change rapidly.
- DevOps-mature organizations can pivot quickly, respond to competitive threats, and capitalize on emerging opportunities faster than competitors.

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# Consider two contrasting scenarios:

- **Company A:** Deploys monthly with numerous quality issues, spends 60% of engineering time on maintenance, and loses key talent due to frustration with rigid processes.

- **Company B:** Deploys daily with minimal issues, spends 80% of time on innovation, and attracts top engineers who value their modern practices.

- Which organization will dominate the market by 2026?

# Consider two contrasting scenarios:

- The business case becomes clear: DevOps transformation isn't about engineering preferences—it's about creating the technical capabilities necessary for business growth, innovation, and competitive advantage.

- For engineering leaders in particular, successful DevOps transformation represents one of the highest-impact contributions you can make to organizational success.

- It directly connects technical excellence to business outcomes in a measurable, undeniable way.

# The 10 essential steps to a high-impact DevOps transformation

- DevOps transformation isn't a linear journey; it's iterative, courageous, and requires strategic leadership.
- These ten steps provide a framework that balances technical changes with the equally critical people and process components.
- **1. Baseline current workflows, bottlenecks, and team sentiment**
- Before making changes, establish a clear picture of your current state.
- This requires both quantitative and qualitative assessment:

# The 10 essential steps to a high-impact DevOps transformation

- **Measure your current DORA metrics** to establish a performance baseline across deployment frequency, lead time, change failure rate, and time to restore.
- **Map your value streams** to visualize how work flows from idea to production, identifying handoffs, wait states, and approval gates that create bottlenecks.
- **Conduct anonymous developer surveys** to understand team pain points, perceived barriers, and cultural readiness for change.

# The 10 essential steps to a high-impact DevOps transformation

- **Inventory your existing toolchain** to identify fragmentation, manual steps, and integration challenges.

- This baseline serves two critical purposes: it identifies your highest-impact improvement opportunities and provides clear before/after metrics to demonstrate transformation success.

# The 10 essential steps to a high-impact DevOps transformation

- **2. Define a vision that ties engineering goals to business outcomes**

- Effective DevOps transformation needs a compelling vision that connects technical practices to business value. Your vision should:
- **Articulate how technical improvements drive specific business outcomes** like faster time-to-market, improved customer satisfaction, or increased innovation capacity.
- **Set ambitious but achievable goals** with clear timelines and success metrics.
- **Establish a "north star" for decision-making** that guides prioritization when you face competing options.
- **Use language that resonates with both technical teams and executives** to ensure alignment across organizational layers.
- Your vision becomes the foundation for communication, prioritization, and measuring progress throughout the transformation journey.

# The 10 essential steps to a high-impact DevOps transformation

- **3. Build cross-functional squads focused on product delivery**

- Traditional department boundaries between development, operations, QA, and security create handoffs that slow delivery and diffuse responsibility. Restructure into cross-functional squads that:
- **Align team composition with product or service boundaries** rather than technical specialties.
- **Embed operations, security, and quality engineers** directly within development teams.
- **Grant end-to-end ownership** from development through production support.
- **Establish shared metrics and goals** that reinforce collective responsibility for outcomes.
- This structural reorganization breaks down silos, accelerates decision-making, and creates shared ownership for both speed and stability.

# The 10 essential steps to a high-impact DevOps transformation

- **4. Prioritize psychological safety before tooling changes**

- Technical changes fail without a foundation of psychological safety: the confidence that team members can take risks without fear of blame. Before implementing new tools or processes:
- **Establish blameless postmortems** that focus on system improvements rather than individual errors.
- **Recognize and reward learning from failure** to encourage experimentation.
- **Create forums for honest feedback** where teams can voice concerns without repercussion.
- **Model vulnerability as a leader** by acknowledging mistakes and demonstrating a growth mindset.
- Psychological safety enables the transparency, experimentation, and continuous learning essential for DevOps success.

# The 10 essential steps to a high-impact DevOps transformation

- **5. Establish CI/CD pipelines and testing gates**

- With the foundational cultural elements in place, begin building the technical infrastructure that enables rapid, reliable delivery:
- **Implement continuous integration** with automated build verification testing to detect issues early.
- **Develop deployment pipelines** that standardize the path to production across teams.
- **Automate security scanning and compliance checks** to shift security left without slowing delivery.
- **Create progressive testing strategies** that balance speed with quality assurance.

- Rather than viewing CI/CD as a purely technical implementation, position it as enabling infrastructure that supports your teams' need for speed, feedback, and quality.

# The 10 essential steps to a high-impact DevOps transformation

- **6. Automate infrastructure provisioning and rollback plans**

- Infrastructure automation creates consistency, minimizes human error, and enables rapid recovery when issues emerge:
- **Implement infrastructure as code (IaC)** to manage environments through version-controlled configuration rather than manual processes.
- **Establish golden paths** for provisioning that incorporate security and compliance requirements by default.
- **Create automated rollback mechanisms** for rapid recovery when deployments fail.
- **Standardize environments** across development, testing, and production to eliminate "works on my machine" problems.

- Automating infrastructure changes creates reproducibility and safety nets that give teams confidence to deploy more frequently.

# The 10 essential steps to a high-impact DevOps transformation

- **7. Choose an integrated DevOps toolchain — not a "Frankenstack“**

- Tool fragmentation creates friction that slows delivery and frustrates teams.
- Build a cohesive toolchain that:
- **Prioritizes integration capabilities** over individual feature richness.
- **Minimizes context switching** by leveraging tools that work well together.
- **Balances standardization with flexibility** to meet unique team needs.
- **Considers the entire development lifecycle** from planning through monitoring.

- Your toolchain choices should reduce cognitive load for developers rather than adding complexity through poor integrations.

# The 10 essential steps to a high-impact DevOps transformation

- **8. Measure everything: DORA, engagement, lead time for learning**

- Create a comprehensive measurement framework that balances delivery performance with team health:
- **Track DORA metrics** to measure technical performance improvements.
- **Monitor team engagement** through regular pulse surveys and 1:1 conversations.
- **Measure "lead time for learning"** ( how quickly you gather feedback on new features).
- **Connect technical metrics to business outcomes** to demonstrate transformation value.

- Effective metrics serve as navigation instruments that guide your transformation journey and demonstrate progress to stakeholders.

# The 10 essential steps to a high-impact DevOps transformation

- **9. Launch a transformation pilot with clear success criteria**

- Rather than attempting a big-bang transformation, start with a focused pilot:
- **Select a team with the right mix of challenges and readiness** to demonstrate transformation value.
- **Define clear success criteria** that blend technical, cultural, and business outcomes.
- **Time-box the pilot** to maintain focus and urgency.
- **Provide additional support and resources** to ensure pilot success.

- A well-executed pilot creates momentum, proves concepts, and builds organizational confidence in your transformation approach.

# The 10 essential steps to a high-impact DevOps transformation

- **10. Commit to quarterly retrospectives and public wins**

- Transformation is iterative, requiring regular reflection and visible success stories:
- **Conduct quarterly retrospectives** to evaluate progress, capture learnings, and adjust your approach.
- **Celebrate and publicize wins** to build momentum and demonstrate value.
- **Share lessons learned** openly across the organization to accelerate adoption.
- **Adjust roadmaps based on feedback** to address emerging challenges and opportunities.

- This commitment to reflection and adaptation prevents your transformation from becoming rigid or disconnected from evolving needs.

# Building Competencies: Full Stack Developers in DevOps

- In today's rapidly evolving software landscape, organizations demand professionals who can seamlessly bridge development and operations.
- **Full Stack Developers with DevOps competencies** are uniquely positioned to deliver end-to-end solutions — from coding the application to automating deployments, ensuring scalability, reliability, and continuous delivery.

# Building Competencies: Full Stack Developers in DevOps

- **2. Core Competencies Required**
- **a) Technical Skills**
- **Frontend Development:** HTML5, CSS3, JavaScript frameworks (React, Angular, Vue).
- **Backend Development:** Node.js, Python (Flask/Django), Java (Spring Boot), PHP.
- **Databases:** Relational (MySQL, PostgreSQL) and NoSQL (MongoDB, Redis).
- **Version Control:** Git/GitHub/GitLab for collaboration.

# Building Competencies: Full Stack Developers in DevOps

- **b) DevOps Skills**
- **CI/CD Pipelines:** Jenkins, GitHub Actions, GitLab CI for automated testing and deployment.
- **Containerization:** Docker for application packaging.
- **Orchestration:** Kubernetes or Docker Compose for scaling and resilience.
- **Infrastructure as Code (IaC):** Terraform, Ansible, or CloudFormation.
- **Cloud Platforms:** AWS, Azure, GCP for hybrid and multi-cloud deployments.
- **Monitoring & Logging:** Prometheus, Grafana, ELK Stack for observability.

# Building Competencies: Full Stack Developers in DevOps

- **c) Soft Skills**
- **Collaboration:** Agile/Scrum methodology participation.
- **Problem Solving:** Debugging across the stack and resolving infrastructure bottlenecks.
- **Adaptability:** Keeping up with evolving DevOps tools and cloud-native practices.

# Building Competencies: Full Stack Developers in DevOps

- **3. Learning Roadmap**
- **Foundations:** Core programming + database skills.
- **System Design & APIs:** RESTful and GraphQL design.
- **DevOps Fundamentals:** CI/CD pipelines, Git workflows.
- **Cloud Native Development:** Deploying full-stack apps on AWS/Azure/GCP.
- **Advanced DevOps:** Microservices, container orchestration, and serverless computing.
- **Security (DevSecOps):** Secure coding, vulnerability scanning, compliance checks.

# Building Competencies: Full Stack Developers in DevOps

- **4. Role in the Industry**
- **Accelerating Delivery:** Ability to develop, test, and deploy quickly with automation.
- **Bridging Silos:** Connecting development and IT operations seamlessly.
- **Scaling Systems:** Designing fault-tolerant, cloud-ready applications.
- **Career Growth:** High demand in startups, enterprises, and cloud service providers.

# Building Competencies: Full Stack Developers in DevOps

- A **Full Stack Developer with DevOps competencies** is not just a coder, but an architect of resilient, automated, and scalable digital solutions.

- By mastering both development and operations, such professionals drive **digital transformation, faster innovation, and reliable software delivery**.

# Self-Organized Teams in DevOps

- 1. DevOps emphasizes **collaboration, automation, and continuous delivery**.
- At the heart of this culture are **self-organized teams** — groups empowered to make decisions, manage workflows, and deliver business value with minimal managerial intervention.

- **2. Key Characteristics of Self-Organized Teams**
- **Autonomy:** Teams decide how to achieve goals instead of being micromanaged.
- **Shared Responsibility:** Development, testing, deployment, and operations are everyone's responsibility.
- **Cross-Functionality:** Each team has diverse skills — developers, testers, operations engineers, and sometimes business analysts — enabling end-to-end delivery.
- **Continuous Learning:** Teams embrace experimentation, feedback, and adaptation.
- **Accountability:** Members hold themselves accountable for outcomes, not just tasks.

# Self-Organized Teams in DevOps

- **3. Benefits in a DevOps Environment**

- **Faster Delivery:** Teams own the entire software lifecycle, reducing handoff delays.

- **Improved Quality:** Continuous integration, testing, and monitoring reduce defects.

- **Higher Innovation:** Teams can experiment and adapt processes/tools without waiting for approval.

- **Employee Engagement:** Autonomy increases motivation and ownership.

- **Resilience:** Teams can adapt quickly to failures, incidents, or changing business priorities.

# Self-Organized Teams in DevOps

- **4. Practices that Enable Self-Organization**

- **Clear Goals & Vision:** Teams must understand business objectives.

- **Automation:** CI/CD, Infrastructure as Code (IaC), and monitoring tools reduce manual overhead.

- **Agile Practices:** Scrum or Kanban help teams self-manage work.

- **Feedback Loops:** Daily standups, retrospectives, and monitoring dashboards support rapid learning.

- **Trust & Empowerment:** Leadership trusts teams to make decisions and gives them authority to act.

# Self-Organized Teams in DevOps

- **5. Example Scenario**
- A **self-organized DevOps team** working on an e-commerce platform:
- **Developers** push code into GitHub.
- **CI/CD pipelines** (Jenkins/GitHub Actions) automatically build, test, and deploy to a staging environment.
- **Operations engineers** monitor performance using Prometheus and Grafana.
- The **team collectively reviews incidents** and implements fixes without needing managerial approval.
- This approach shortens the release cycle, improves reliability, and boosts customer satisfaction.

# Self-Organized Teams in DevOps

- In DevOps, **self-organized teams are the backbone of continuous delivery and innovation**.
- By blending cross-functional expertise, automation, and autonomy, these teams deliver high-quality software faster while remaining adaptable to dynamic business needs.

# Traditional Teams vs. Self-Organized Teams in DevOps

| Aspect | Traditional Teams | Self-Organized DevOps Teams |
|---|---|---|
| **Decision Making** | Manager-driven, top-down | Team-driven, decentralized |
| **Work Distribution** | Assigned by manager | Shared ownership and collaborative task selection |
| **Skills** | Specialized (e.g., only dev, only ops) | Cross-functional (dev, ops, QA, monitoring) |
| **Responsibility** | Individuals responsible for tasks | Team collectively responsible for outcomes |
| **Adaptability** | Slower to react to changes | Highly adaptive; continuous improvement |
| **Innovation** | Limited, dependent on approvals | High; empowered to experiment and iterate |
| **Feedback Loops** | Long cycles (monthly/quarterly reviews) | Short, continuous (standups, retrospectives, metrics) |
| **Delivery Speed** | Slower due to handoffs and silos | Faster with CI/CD and automation |
| **Motivation & Ownership** | Lower; task-focused mindset | Higher; goal-oriented and value-driven |

# Intrinsic Motivation in DevOps

- In a DevOps culture, **people are as important as processes and tools**.

- While automation and CI/CD pipelines streamline technical tasks, the real success of DevOps comes from the **intrinsic motivation** of team members — their inner drive to learn, collaborate, and deliver value beyond external rewards.

- **2. What is Intrinsic Motivation?**

- Intrinsic motivation refers to the **internal desire to perform a task** because it is personally rewarding, not because of external pressures like salary, deadlines, or recognition.

- In DevOps, this means individuals take ownership of quality, reliability, and innovation **because they care about the outcome**.

# Intrinsic Motivation in DevOps

- **3. Key Drivers of Intrinsic Motivation in DevOps**

- **Autonomy** – The freedom to make decisions about tools, methods, and workflows.

- **Mastery** – The opportunity to continuously improve technical and problem-solving skills.

- **Purpose** – A clear understanding of how their work impacts the business, customers, and society.

- **Collaboration** – Working in cross-functional teams where knowledge-sharing and peer learning are valued.

- **Trust & Empowerment** – Teams feel trusted by leadership to experiment, take risks, and recover from failures.

# Intrinsic Motivation in DevOps

- **4. Benefits of Intrinsic Motivation in DevOps**

- **Higher Quality Work:** Motivated teams naturally pay attention to detail and proactively fix issues.

- **Innovation:** Passion for solving problems leads to creative approaches and new solutions.

- **Resilience:** Intrinsically motivated members see failures as learning opportunities.

- **Employee Retention:** Engaged individuals stay longer and contribute more deeply.

- **Continuous Improvement:** Teams drive adoption of new tools, practices, and automation without external enforcement.

# Intrinsic Motivation in DevOps

- **5. Example Scenario**
- In a healthcare DevOps team:
- Developers **implement automated testing** not just because management demands it, but because they want to **ensure patient data safety and system reliability**.

- Operations engineers **proactively monitor system performance** and fine-tune resources because they want to **make sure the hospital can access life-saving data without delay**.

- This sense of **purpose and responsibility** fuels continuous improvement.

- Intrinsic motivation is the **heartbeat of DevOps culture**. By fostering autonomy, mastery, and purpose, organizations can empower teams to deliver high-quality software faster, innovate continuously, and build sustainable, resilient systems.

# Technology in DevOps

- **2. Infrastructure as Code (IaC)**

- **Definition:** IaC is the practice of managing and provisioning infrastructure (servers, networks, storage) using code instead of manual processes.

- **Tools:** Terraform, Ansible, AWS CloudFormation, Puppet, Chef.

- **Benefits:**
  - Consistency across environments (Dev, QA, Prod).
  - Version-controlled infrastructure, enabling rollbacks.
  - Scalability through automation (auto-scaling clusters, VMs, containers).

- **Example:** Deploying a Kubernetes cluster automatically with Terraform scripts.

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# Technology in DevOps

- **3. Delivery Pipeline**
- **Definition:** A **delivery pipeline** automates the software build, test, and deployment process, ensuring that code changes move smoothly from development to production.
- **Stages:**
  - **Source Code Management** (Git/GitHub/GitLab).
  - **Continuous Integration (CI):** Automated builds and unit testing (Jenkins, GitHub Actions, GitLab CI/CD).
  - **Continuous Delivery (CD):** Automated deployment to staging/production environments.
- **Benefits:**
  - Faster release cycles.
  - Early detection of defects.
  - Reduced manual intervention.
- **Example:** A Jenkins pipeline that builds a Docker image, runs unit/integration tests, and deploys to AWS EC2.

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# Technology in DevOps

- **4. Release Management**
- **Definition:** Release Management in DevOps involves planning, scheduling, and controlling software delivery into production environments.
- **Approach in DevOps:**
  - **Continuous Release:** Frequent small updates rather than big-bang releases.
  - **Blue-Green Deployment:** Two environments (blue and green) are used to minimize downtime.
  - **Canary Releases:** Gradually releasing updates to a subset of users before full rollout.
- **Benefits:**
  - Minimized risks during deployment.
  - Improved stability with automated rollbacks.
  - Better alignment with business needs.
- **Example:** Using Kubernetes + Helm charts for rolling updates with zero downtime.

# Technology in DevOps

- Technologies like **IaC, Delivery Pipelines, and Automated Release Management** form the backbone of DevOps.
- They ensure **reliable, repeatable, and scalable deployments**, enabling organizations to deliver high-quality software faster and with greater confidence.

# Tools and Technologies as Enablers for DevOps

- DevOps is built on a culture of collaboration, automation, and continuous improvement.
- However, the **real enablers of DevOps success are the tools and technologies** that make it possible to achieve speed, reliability, and scalability.
- These tools span the entire software development lifecycle (SDLC), from code integration to monitoring in production.

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# Key Categories of DevOps Tools

- **a) Version Control**
- **Purpose:** Manage code, track changes, enable collaboration.
- **Examples:** Git, GitHub, GitLab, Bitbucket.
- **Enabler Role:** Facilitates team collaboration and code traceability.

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# Key Categories of DevOps Tools

- **b) Continuous Integration & Continuous Delivery (CI/CD)**
- **Purpose:** Automate build, test, and deployment pipelines.
- **Examples:** Jenkins, GitLab CI/CD, GitHub Actions, CircleCI, Bamboo.
- **Enabler Role:** Ensures faster, error-free deployments with minimal manual intervention.

# Key Categories of DevOps Tools

- **c) Infrastructure as Code (IaC) & Configuration Management**
- **Purpose:** Automate provisioning and configuration of infrastructure.
- **Examples:** Terraform, Ansible, Puppet, Chef, AWS CloudFormation.
- **Enabler Role:** Provides consistency, scalability, and version-controlled infrastructure.

# Key Categories of DevOps Tools

- **d) Containerization & Orchestration**
- **Purpose:** Package applications with dependencies and run them anywhere.
- **Examples:** Docker, Kubernetes, OpenShift, Docker Compose.
- **Enabler Role:** Supports microservices, portability, and high availability.

# Key Categories of DevOps Tools

- **e) Monitoring & Logging**
- **Purpose:** Ensure system reliability and observability.
- **Examples:** Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Splunk.
- **Enabler Role:** Enables proactive issue detection, performance optimization, and feedback loops.

# Key Categories of DevOps Tools

- **f) Collaboration & Communication**
- **Purpose:** Align development, operations, and business teams.
- **Examples:** Slack, Microsoft Teams, Jira, Confluence, Trello.
- **Enabler Role:** Promotes transparency, agility, and effective communication.

# Key Categories of DevOps Tools

- **3. Benefits of Tools as DevOps Enablers**
- **Automation:** Eliminates repetitive manual tasks.
- **Consistency:** Ensures uniform environments across Dev, QA, and Prod.
- **Scalability:** Rapidly scale applications in cloud or hybrid environments.
- **Reliability:** Continuous monitoring improves uptime and user experience.
- **Faster Delivery:** Shorter release cycles enable quicker time-to-market.

# Key Categories of DevOps Tools

- **4. Example Scenario**
- A fintech company uses:
- **GitHub** for source control,
- **Jenkins** for CI/CD,
- **Docker & Kubernetes** for microservices deployment,
- **Terraform** for infrastructure automation,
- **Prometheus & Grafana** for monitoring.
- This integrated toolchain enables them to **release updates multiple times a day**, reduce downtime, and ensure compliance with financial regulations.

- DevOps tools are not just add-ons; they are the **backbone of the DevOps ecosystem**.

- When strategically chosen and integrated, they empower organizations to achieve **continuous delivery, agility, and business value**.

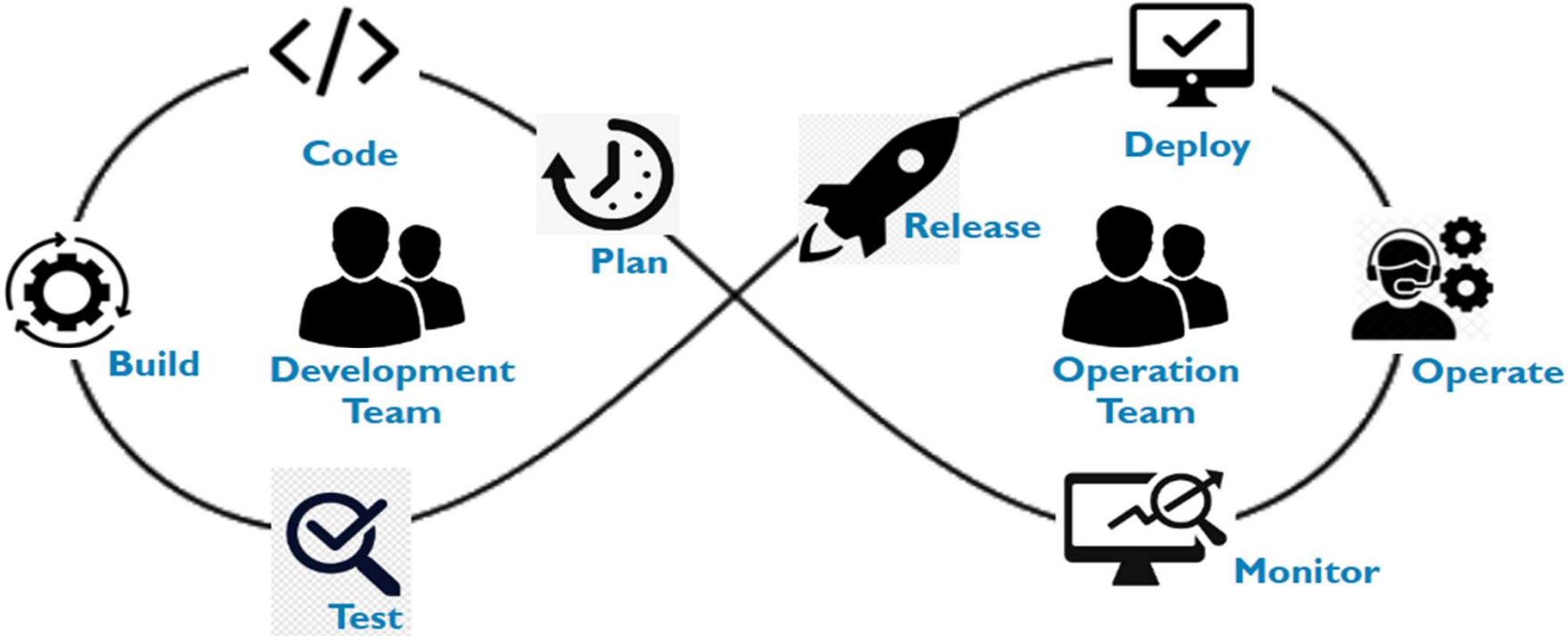
# Development with DevOps

Streamlined Deliveries

Continuous Monitoring and Feedback

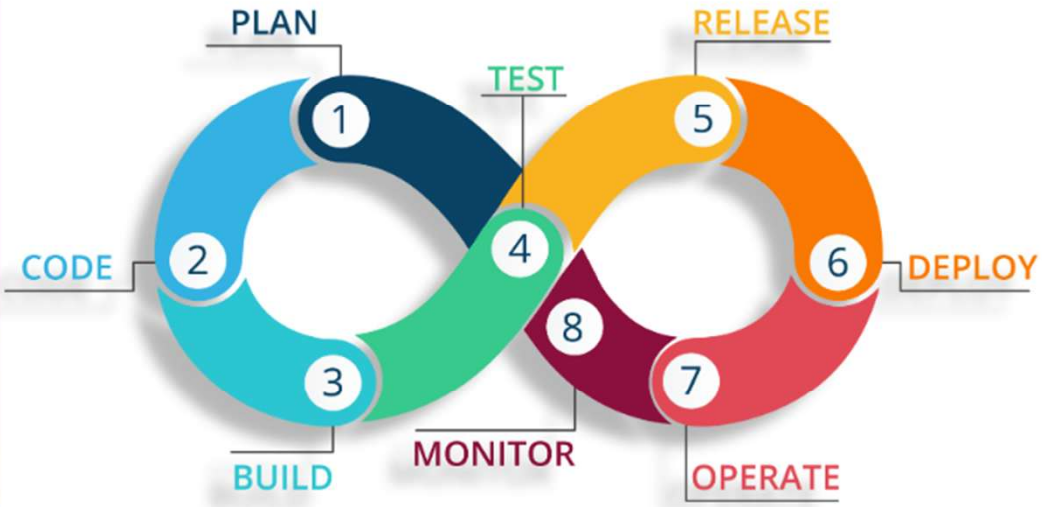
Team Work in Collaboration

# DevOps in Real Life



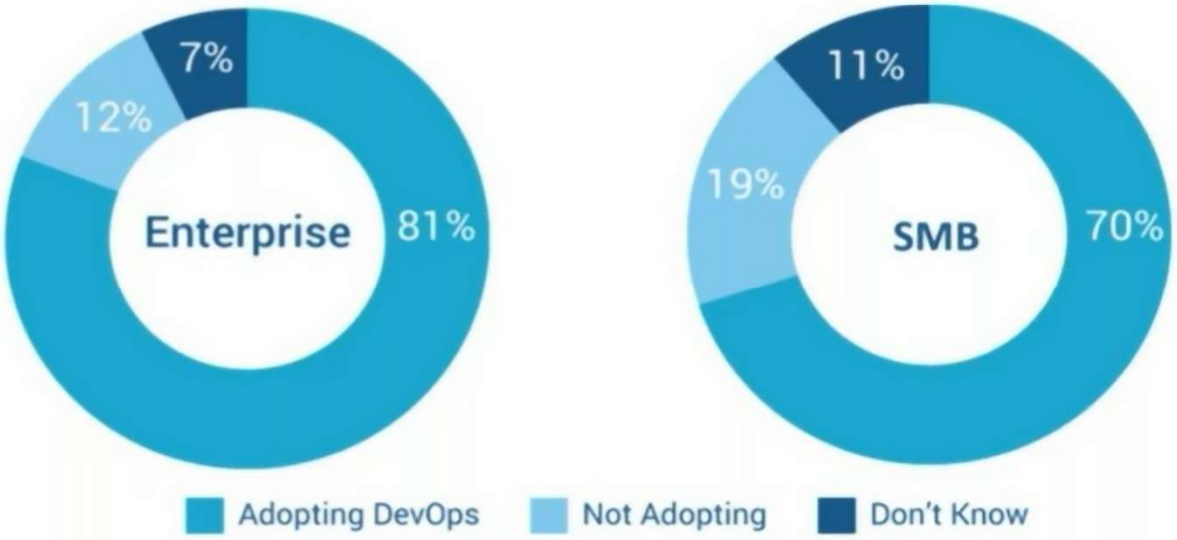
# DevOps in Real Life

- Integrates developers and operations teams
- Improves collaboration and productivity by:
  - ✓ Automating infrastructure
  - ✓ Automating workflows
  - ✓ Continuously measuring application performance



# DevOps Current Scenario

"Considering the changing pace of IT landscape, almost all the companies require fast paced development environment"



# DevOps Lifecycle - Plan

"First stage of DevOps cycle, where you Plan, Track, Visualize and Summarize your Project before working/starting it."

**Planning Tools**

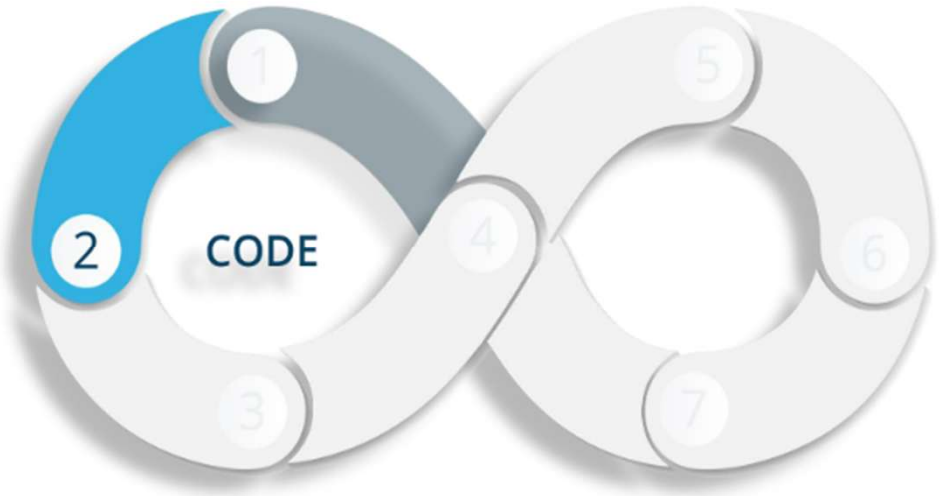
JIRA

Trello

Tricentis

# DevOps Lifecycle - Code

"Second stage of DevOps cycle, where the developers write their code"




**Coding Tools**

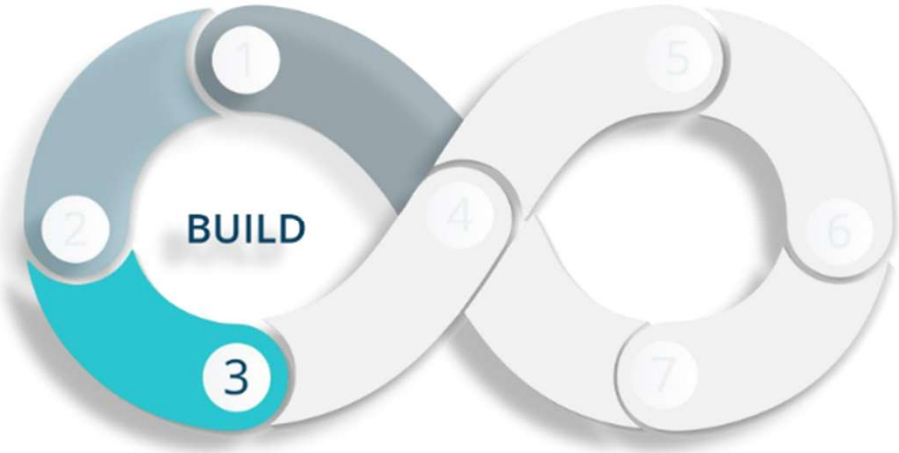
eclipse

git

SUBVERSION

# DevOps Lifecycle - Build

"Build is a pre-release version and is identified by a build number, rather than by a release number"




**Building Tools**

- Apache Ant
- Jenkins
- Maven
- gradle — a better way to build

# DevOps Lifecycle - Test

"Process of executing automated tests as part of the software delivery pipeline in order to obtain feedback on the business risks associated with a software release as rapidly as possible"

**Testing Tools**

APACHE JMeter™

JUnit

Se

# DevOps Lifecycle - Release

"This phase helps to integrate code into a shared repository using which, you can detect and locate errors quickly and easily"

**Releasing Tools**

- Travis CI
- GitLab
- Bamboo

# DevOps Lifecycle - Deploy

"Manage and maintain development and deployment of software systems and servers in any computational environment"

**Deploying Tools**

aws

ANSIBLE

CHEF

# DevOps Lifecycle - Operate

"This phase is to keep the system upgraded with the latest update"

**Operating Tools**

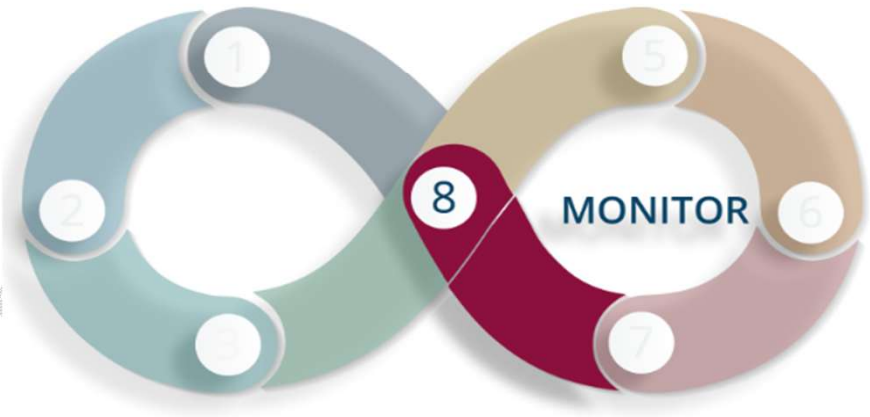
aws

ANSIBLE

CHEF

# DevOps Lifecycle - Monitor

"It ensures that the application is performing as desired and the environment is stable. It quickly determines when a service is unavailable and understand the underlying causes"



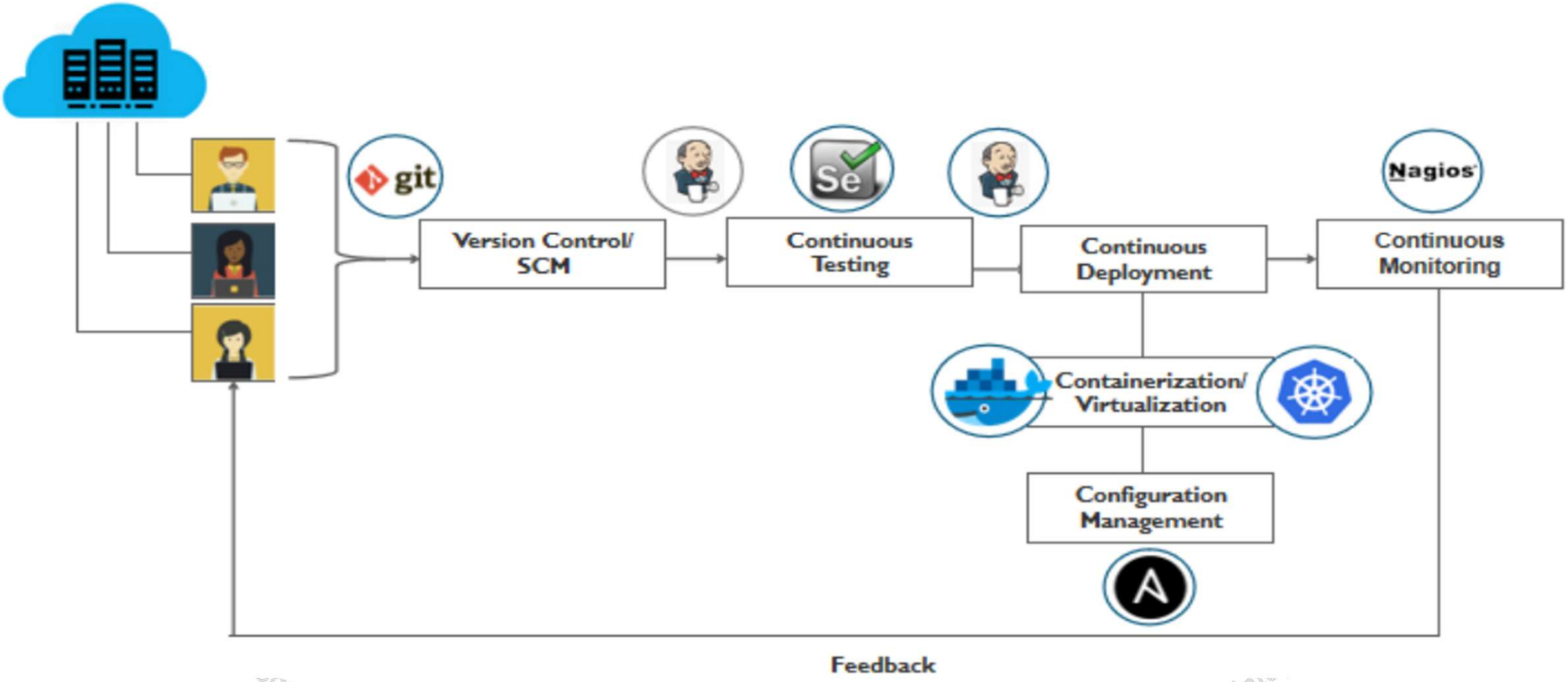

**Monitoring Tools**

Nagios

sensu

splunk>

# DevOps Stages



# DevOps Stages

**Continuous Integration (CI):**

- ✓ Continuous integration is a DevOps software development practice where developers regularly merge their code changes into a central repository, after which automated builds and tests are run.
- ✓ The key goals of continuous integration are to find and address bugs quicker, improve software quality, and reduce the time it takes to validate and release new software updates.

**Continuous Delivery (CD):**

- ✓ CD ensures that code is always in a state ready for deployment, streamlining release processes and reducing risks.
- ✓ With continuous delivery, code changes are automatically built, tested, and prepared for a release to production.
- ✓ Continuous delivery expands upon continuous integration by deploying all code changes to a testing environment and/or a production environment after the build stage.

**Continuous Deployment (CD):**

- ✓ CD automates the release to production, increasing deployment frequency while maintaining high standards of reliability.

# QUESTIONS AND DISCUSSION

# THANK YOU

Work Integrated Learning Programmes